

Muhammad Daniyal

BS Computer Science '27 | Full-Stack & AI Engineering

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PROFILE

CS undergraduate with experience in AI-powered applications, full-stack development, and performance optimization. Skilled in building scalable web systems, machine learning pipelines, and parallel computing solutions using modern development frameworks and GPU acceleration technologies.

PROFESSIONAL EXPERIENCE

Software Engineer 01 Aug 2026 — Present
Avants Lab · Remote

- Developing and maintaining dynamic web applications and mobile solutions, collaborating with cross-functional teams to deliver user-centric features.
- Participating in the full software development lifecycle, from requirements gathering to deployment and iteration.

Software Engineer Intern 15 Jun 2026 — 01 Aug 2026
Avants Lab · Remote

- Built and shipped features across web and mobile products alongside cross-functional teams.
- Converted to a full-time Software Engineer at the end of the internship.

Cybersecurity Intern Jun 2025 — Aug 2025
NUST Agristreams · Islamabad, Pakistan

- Implemented backend security monitoring features for a host-based WAF using Python.
- Developed real-time log analysis and monitoring dashboards using Flask.
- Worked with rule-based request inspection and attack detection workflows.

EDUCATION

BS Computer Science Expected 2027
FAST (NUCES) · Islamabad, Pakistan

- Relevant coursework:** Data Structures & Algorithms, Database Systems, Computer Networks, Computer Architecture, High Performance Computing, Software Design & Engineering, Parallel & Distributed Computing, Artificial Intelligence.

SKILLS

Languages C/C++, Python, Java, SQL, JavaScript, TypeScript
Frameworks & Web React, Next.js, Node.js, Express, MongoDB, Flask, Streamlit, Tailwind
Parallel & Systems CUDA, OpenMP, OpenACC, OpenMPI, OpenCL, AVX2 SIMD, Linux
Tools & Platforms Git, GitHub, perf, Scikit-learn

PROJECTS

AI Chat & Booking SaaS Aug 2026
Next.js, TypeScript, MongoDB, Groq LLM

- Built a multi-tenant AI chatbot SaaS that embeds on external websites via Shadow DOM and iframe, enabling real-time customer chat, automated booking management, and natural-language availability checking.
- Engineered a hybrid 2-step AI architecture that injects live database state into LLM calls to eliminate hallucinations for calendar availability queries.
- Implemented atomic usage tracking with MongoDB *\$inc*, TTL-based data lifecycle management, *\$bsonSize* storage aggregation, and strict CORS domain whitelisting for cross-origin widget security.

Parallel K-Means Optimization May 2026
C++, OpenMP, AVX2 SIMD Intrinsics, perf

- Optimized K-Means clustering using OpenMP parallelization, SIMD vectorization, and cache-aware memory layouts, achieving up to **10x speedup** over the baseline implementation.

Benchmarked scalability, convergence behavior, and workload performance across multiple optimized implementations.	
KLT Feature Tracking Optimization <i>CUDA, OpenACC, Computer Vision</i>	Nov 2025
- Accelerated KLT feature tracking using CUDA and OpenACC, achieving 2x speedup over the CPU baseline. - Benchmarked CPU, OpenACC, and CUDA implementations with focus on execution time, memory access patterns, and GPU parallelization strategies.	
Treap vs BST Performance Analysis <i>C++, Data Structures, Algorithms, Python</i>	Nov 2025
- Implemented and benchmarked Treap and BST data structures on a large-scale Reddit dataset (> 10,000 operations). - Achieved 56x faster insertion and search than BST while analyzing balancing behavior and tree height differences (25 vs 4025 levels). - Visualized timing and structural metrics in Python to demonstrate empirical $O(\log n)$ vs $O(n)$ behavior.	
PetCare <i>Next.js, TypeScript, MongoDB</i>	May 2026
- Developed a full-stack pet care and e-commerce platform featuring authentication, product management, cart/checkout, bookings, and admin dashboards. - Implemented secure role-based workflows for customers, veterinarians, and administrators.	
AI-Powered Quiz Generation System <i>Python, Scikit-learn, Streamlit</i>	May 2026
- Built an automated quiz generation system using TF-IDF features and a 4-model ensemble (Logistic Regression, Naive Bayes, SVM, KMeans) for question generation and answer evaluation. - Developed a real-time Streamlit interface with progressive hints, difficulty-ranked distractors, and live performance tracking.	

LEADERSHIP & ACTIVITIES

Research & Development Society (RnD)	FAST LADS
Active member participating in research-oriented technical activities and discussions.	
Outreach & Engagement Society	Meri Kahani
Web Development Team member supporting digital initiatives and outreach activities.	
Orphanage Muhammadi Aghosh	Social Worker
Organized educational activities and fundraising initiatives for orphaned children.	